CS331 Accessibility Report

Computer Security and Information Assurance, Spring 2027

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Summary

CS331 meets WCAG 2.2 Level AA, which includes every WCAG 2.1 AA criterion, on the course website and in the Canvas master shell. It also meets 25 of the 31 Level AAA criteria. Of the other 6, 5 do not apply because the course has no video or audio, and the 1 that is not met is listed under Level AAA with the reason.

The first pass found AA failures on every website page and on 4 Canvas pages, and a second pass at AAA found more. All of them are fixed, and the same checks now find none. I reran them on October 28, after the site's theme was restyled, and they still find none. The textbook PDF (Portable Document Format) was the biggest gap, so every chapter the course assigns is now a web page on the course site that passes the same AAA checks (see CyBOK textbook). Three items are outside what I can fix in the course itself. They are listed under Known limitations with how a student gets an accessible version.

Check	Level	Before	After
Website pages with an axe-core failure (light or dark mode)	AA	67 of 67	0
Canvas bodies with an axe-core failure	AA	4 of 77	0
Contrast under 4.5:1, worked out by hand	AA	1 (the schedule caret)	0
Link text that is unclear or points two places	AA	1	0
Pages that do not reflow at 320 px	AA	0	0
Tab stops with no visible focus	AA	0	0
Website pages with text under 7:1 contrast	AAA	67 of 67	0
Controls under 44 by 44 px	AAA	every page	0

Check	Level	Before	After
External links that open a new tab without saying so	AAA	every external link	0
Paragraphs closer than 1.5 lines apart	AAA	7	0
Images of text	AAA	1 (the home page banner)	0
Problems UDOIT reports in Canvas (errors and suggestions)		398, then 55 more	0, plus the long CyBOK chapters (see below)
Problems Ally reports in Canvas		20	0, plus 17 CyBOK figures (see below)
Links that go nowhere (a missing page, anchor or a 404)	AA	46	0
Assigned CyBOK chapters readable without the un-tagged PDF	AAA	0 of 15	15 of 15

Scope

- **Website**: all 83 pages under https://shanep.github.io/cs331/. That is the syllabus and the other top-level pages, the schedule, 30 module overview and notes pages, 11 labs, 6 discussions, the Course Resources, reminder and data pages, and 16 CyBOK pages (the 15 assigned chapters and their index). The site theme fixes below apply to every course on the site, not only CS331.
- **Canvas**: all 77 bodies in the CS331 master shell (Canvas course 52107): pages, assignments, discussions, the survey, and the syllabus. Canvas's own interface (navigation, menus, buttons, the rich content editor) belongs to Instructure and is not part of this review.
- **Files**: the one document students download, the CyBOK v1.2.0 PDF.

How I checked

Every check runs from one command in the course repository, so anyone can rerun it and get the same numbers:

```
npm run a11y -- --course cs331 --level aaa --canvas <edutools pull snapshot>
```

It builds the site, then runs:

1. **axe-core 4.14.0** (Deque) in Chromium through Playwright 1.57.0, with the `wcag2a`, `wcag2aa`, `wcag21a`, `wcag21aa`, `wcag22aa`, `wcag2aaa` and `best-practice` rule sets, on every page in light

mode and again in dark mode.

2. **Contrast by hand** for every element axe marks “needs review” because it cannot tell the background (text over a layered background, mostly). The script reads the text color and the first opaque background behind it and computes the ratio with the WCAG formula. At AAA that is 7:1 for text and 4.5:1 for large text.
3. **Link text** (2.4.4, 2.4.9): no “click here” or “read more”, no raw URLs as link text, no text that points to two different places on one page, and no link that opens a new tab without saying so in its name (3.2.5).
4. **Reflow** (1.4.10): every page at 320 CSS px wide, the width of a 1280 px screen at 400% zoom. A page fails if it scrolls sideways. Tables, code blocks and display math may scroll inside their own box, which WCAG allows.
5. **Focus** (2.4.7, 2.4.12, 2.4.13): tab through every page and flag any stop with no focus ring, a ring thinner than 2 px or under 3:1, or any part hidden under the sticky navigation bar.
6. **Target size** (2.5.5): every control at 1280 px and at 320 px, measured by where a click actually lands, must take a click across at least 44 by 44 px. Links inside a paragraph, list item or table cell are exempt, as the criterion allows for targets in a block of text. Footnote links count as part of their sentence.
7. **Text spacing and images of text** (1.4.8, 1.4.9): line spacing of at least 1.5, paragraphs at least 1.5 lines apart, no justified text, and no image whose alt text is a title.
8. **Canvas**: the same axe rules and link checks on every body in a fresh `edutools pull` of course 52107, framed the way Canvas shows it (the item title as the page heading, Canvas’s text color on white).
9. **UDOIT**: Boise State’s UDOIT tool in Canvas, scanned on October 28 and 29 with its default filters off, so unpublished items count too. The check now runs the same rules on every website page and Canvas body, so a new problem fails the check before UDOIT ever sees it: a paragraph that is all bold, two links in a row to the same place, alt text over 150 characters, colored text that is not bold or italic, and embedded frames.
10. **Links**: every link in a page body, on the website and in Canvas, has to land somewhere. A link inside the course has to reach a page that exists and an anchor on it, and an outside link fails on a 404 or a 410. An outside site that turns the check away, times out or does not resolve is listed to open by hand, since many sites refuse scripts and my network’s DNS filter blocks a few live ones (petsymposium.org, suricata-ids.org, tsfdn.org). A Canvas link into another course fails too, since Ally checks links without a login.
11. **Ally**: the Accessibility Report in Canvas, read on October 29. The check now fails on the kind of Ally problem it can see, a link into another course. Ally’s image contrast rule reads the pixels of each image, and its image description rule wants a description stored in Ally for each image file, so the check cannot copy either one.

The last run, on October 29, 2026, printed the lines below. It covers the CyBOK pages on the website and in Canvas.

```
cs331 (WCAG 2.2 AAA): 84 site pages in light and dark mode, 94 Canvas bodies
     0  axe violations (site)
     0  contrast under the minimum, worked out by hand
     0  link text problems (site)
     0  UDOIT problems (site)
     0  broken links (site)
     0  pages that do not reflow at 320 px
     0  tab stops with a focus problem
     0  targets under 44 by 44 px
     0  text spacing and images of text
     0  axe, UDOIT and link text problems (Canvas)
     0  broken links (Canvas)
   350  needs-review groups from axe (color-contrast-enhanced, color-contrast,
identical-links-same-purpose, th-has-data-cells); contrast ones are checked above
    32  pages over 3000 words, which UDOIT suggests splitting
   303  of 1236 outside links turned the check away, timed out or did not resolve;
open them by hand
```

The Canvas check reads each body without Canvas's own stylesheet, so I also measured the text colors on live Canvas pages (the syllabus, the home page, a notes page, the assignment list and a discussion). Course content is at 7:1 or better everywhere except inline code, which Canvas draws red on gray at 5.21:1. The course stylesheet now sets inline code to the page text color.

I also checked by hand what a script cannot judge:

- **Images**: the 215 module icons have empty alt text, because they repeat the heading next to them. The home page banner is now decoration only (see below), and the instructor photo has real alt text. Canvas drops an empty alt when it draws a page unless the image also has `role="presentation"`, so every decorative image carries both, and the check now flags one that does not.
- **Instructions** (1.3.3): no step depends on shape, color, or position alone.
- **Diagrams**: the course has no images of text and no ASCII diagrams. Lab 2 asks students for a threat model diagram and accepts a plain-text one, so nobody has to draw.
- **Media**: the course has no video or audio, so captions and transcripts do not apply. The one embed, a Google Doc on the instructor-only page, has a title.
- **Headings and tables**: headings go in order with no skipped levels, and every data table has a header row.
- **Keyboard**: tabbing reaches every link and button with no keyboard trap, and every page has a "Skip to content" link.
- **Time limits**: the one quiz (the survey) has no time limit.
- **Motion**: the site turns off its transitions when the operating system asks for reduced motion.

Level AAA

The W3C does not recommend requiring AAA for a whole site, because some of it cannot be met for every kind of content. Here is where CS331 stands on each criterion.

Criterion	Status	How
1.2.6 Sign Language	Does not apply	No video or audio
1.2.7 Extended Audio Description	Does not apply	No video
1.2.8 Media Alternative	Does not apply	No video
1.2.9 Audio-only (Live)	Does not apply	No live audio
1.3.6 Identify Purpose	Met	Header, navigation, main and aside landmarks on every page, and every icon has a text label
1.4.6 Contrast (Enhanced)	Met	Every color the course sets is at least 7:1, or 4.5:1 for large text
1.4.7 Low or No Background Audio	Does not apply	No audio
1.4.8 Visual Presentation	Met	Light and dark mode, line spacing 1.75, paragraphs at least 1.5 lines apart, no justified text, and the text reflows to any width
1.4.9 Images of Text (No Exception)	Met	None left after the banner fix
2.1.3 Keyboard (No Exception)	Met	Everything works from the keyboard
2.2.3 No Timing	Met	No timed content or quiz
2.2.4 Interruptions	Met	No pop-ups or automatic updates
2.2.5 Re-authenticating	Met	The website has no sign-in, and Canvas sign-in belongs to Boise State
2.2.6 Timeouts	Met	The website keeps no data that could time out
2.3.2 Three Flashes	Met	Nothing flashes
2.3.3 Animation from Interactions	Met	Transitions turn off with reduced motion
2.4.8 Location	Met	The sidebar marks the current page and its module
2.4.9 Link Purpose (Link Only)	Met	Every link names where it goes on its own
2.4.10 Section Headings	Met	Every page is divided by headings

Criterion	Status	How
2.4.12 Focus Not Obscured (Enhanced)	Met	The page scrolls a focused element clear of the sticky bars
2.4.13 Focus Appearance	Met	A 2 px focus ring in the link color, 7:1 or better
2.5.5 Target Size (Enhanced)	Met	Every control outside a block of text is at least 44 by 44 px
2.5.6 Concurrent Input Mechanisms	Met	Nothing limits the input device
3.1.3 Unusual Words	Met	Every notes page has a Key Terms section that defines the terms it uses
3.1.4 Abbreviations	Met	Every abbreviation is spelled out on its first use on each page (see below)
3.1.5 Reading Level	**Not met**	See below
3.1.6 Pronunciation	Met	No word's meaning depends on how it is said
3.2.5 Change on Request	Met	Links open in the same tab, and the two theme links that open a new tab say so
3.3.5 Help	Met	The Course Questions forum, the Technology Support page and the Instructor Information page, all linked from the home page
3.3.6 Error Prevention (All)	Met	The website takes no submissions, and submitting work happens in Canvas's own interface
3.3.9 Accessible Authentication (Enhanced)	Met	The website has no sign-in

3.1.4 Abbreviations. A security course is thick with abbreviations (TLS, RBAC, HMAC, STRIDE and about 90 more). A scan found 98 real ones, used 311 times across the course's pages, and most were never spelled out. Each Canvas page is its own page, so each abbreviation is now spelled out the first time it appears on every page that uses it, as in "TLS (Transport Layer Security)". That changed 54 pages. Words in capitals that are not abbreviations (SPRING BREAK, SHALL NOT), course and room codes, and names like MITRE ATT&CK are left alone.

On the CyBOK pages the book's own wording is kept, so a meaning is added after an abbreviation in brackets, as in "MAC [Message Authentication Code]". The converter picks the meaning per chapter, because the book uses MAC, CSP, TTP and a few others for different things in different chapters, and a note at the top of each page says the bracketed meanings are mine, not the book's.

3.1.5 Reading Level. The criterion asks for text a lower secondary student can read, or a simpler version next to it. CS331 is an upper-division course, and its readings are the CyBOK knowledge areas. The notes and labs are written plainly, but they do not reach that level, and writing a second version of every page is not practical.

What I fixed

Website theme, Level AA (every course)

Problem	WCAG	Where	Fix
The sidebar put a button inside a button on every collapsible section, which screen readers announce badly	4.1.2	670 elements, every page	Override VitePress's sidebar item so a row is a button only when it has no link, and the caret is the button when it does
The search button was named "Search" but showed "Search Ctrl K"	2.5.3	every page	Override the search button so its name comes from its visible text, with the shortcut hidden from screen readers
Code highlighting colors under 4.5:1 (red keywords at 4.23:1 in light mode, gray comments at 3.75:1 in dark mode)	1.4.3	3 pages with code	Switch both code themes to GitHub's high-contrast versions
The code block language label at 2.87:1	1.4.3	every code block	Use the theme's secondary text color
The schedule showed each module title as plain text, and each item's type only as an icon	1.3.1, 1.1.1	schedule page	Make module titles headings, and give each item a hidden text label ("Assignment:", "Page:")
The schedule's decorative caret at 2.87:1	1.4.11	schedule page	Hide it from screen readers and use the secondary text color

Website theme, Level AAA (every course)

Problem	WCAG	Where	Fix
Secondary text at 5.2:1, the link color at 5.8:1, and the tip, warning and danger colors also under 7:1	1.4.6	every page	Darken them in light mode and lighten them in dark mode, to 7:1 or better

Problem	WCAG	Where	Fix
Code comments at 4.66:1 even in the high-contrast themes	1.4.6	every code block	Build the code themes at load time, nudging each token color until it reaches 7:1 on the code background
External links opened in a new tab without warning	3.2.5	every external link	Open them in the same tab, and label the two theme links that still open a tab (the GitHub icon and “Edit this page”)
The browser’s thin focus outline on links	2.4.13	every page	A 2 px solid ring in the link color on everything that takes focus
A focused heading link could scroll under the sticky navigation bar	2.4.12	every page	Scroll padding that clears the bar, and the menu bar on narrow screens
Sidebar rows, outline links, the search button, the theme switch, the GitHub icon, copy buttons, heading anchors and schedule links under 44 px	2.5.5	every page	Make each one 44 px, or widen the area that takes a click where the look should stay the same
No space between paragraphs in a quote, and 8 px in a tip box	1.4.8	5 pages	Space them at least 1.5 lines apart
The sidebar’s “Home” went to the course home page, and the top bar’s “Home” went to the site home page	2.4.9	every course page	Rename the sidebar entry “Course Home”

CS331 content (website and Canvas)

Problem	WCAG	Where	Fix
Tables with an empty header cell	1.3.1	D1 to D6 time tables, 6.01, Module 9, 12.01, 13.01	Name every column (“Activity”, “Subject”, “Defense”)
Two links on the syllabus both read “academic calendar” but went to different pages	2.4.4	syllabus	Name the second one “Spring 2027 academic calendar”

Problem	WCAG	Where	Fix
Lab 6 told students to click “the icon” or “the padlock”, which depends on seeing it	1.3.3	Lab 6 Step 1	Name the control by its accessible name (“the *View site information* button at the left end of the address bar”)
The home page banner drew the course title as a picture	1.4.9	home page	Take the title out of the banner, keep the circuit art as decoration, and let the heading below it carry the title
Canvas draws inline code red on gray at 5.21:1	1.4.6	34 Canvas bodies with code	Set inline code to the page text color in the course stylesheet
Abbreviations were never spelled out	3.1.4	54 pages	Spell out each one on its first use on each page
Three links read “docs/CyBOK_v1.2.0.pdf”, the file’s path	2.4.4	syllabus, Readings and Resources, Lab 0	Name them “CyBOK v1.2.0 (PDF)”

CyBOK textbook

The CyBOK v1.2.0 PDF is 1,115 pages with no tags, so a screen reader gets no headings, tables or reading order. The course assigns parts of 15 of its 22 chapters, so those 15 chapters are now web pages under https://shanepanter.com/cs331/cybok/, and the reading links in every week’s notes go straight to the section on them. The same pages are also Canvas pages, in the CyBOK module, and they link to each other inside Canvas rather than back to the website. The PDF is linked at cybok.org instead of kept as a copy in Canvas.

A script in the repository (`scripts/cybok/convert.py`) builds the pages from the PDF, so they can be rebuilt when CyBOK puts out a new version.

Problem in the PDF	WCAG	Fix on the web pages
No tagged headings, lists or tables	1.3.1	Real headings for every section, real lists, and 351 footnotes that link both ways
Math drawn as glyphs a screen reader cannot read	1.1.1, 1.3.1	MathJax, which draws each formula and hands screen readers a MathML copy
38 figures with no alt text	1.1.1	Alt text for each one, plus a “Text description” that lists everything the figure shows

Problem in the PDF	WCAG	Fix on the web pages
Diagrams and tables drawn as pictures	1.4.9	Rebuilt as real tables where the content is a table
About 1,200 links whose text is a bare web address	2.4.9	Reference links carry the title of the work, and the other 138 addresses get a name from a list in the repository. Made-up addresses in the book's examples are shown as code, not links
Abbreviations never spelled out on most pages	3.1.4	A meaning in brackets on each one's first use on each page

The theme also gained styles for what these pages added: 44 px tall "Text description" toggles and display math, footnote return links of at least 24 px, and inline math that shrinks to fit a phone instead of making the page scroll.

Earlier this month I also fixed the raw URLs Greg Snow flagged in his review of the shell. Every one now sits behind descriptive link text.

Known limitations

- **Ally flags 17 CyBOK figures for image contrast.** The text in them is black on light gray and easy to read, so I kept the book's figures as they are.
- **The CyBOK v1.2.0 PDF is still not tagged.** It is published by the CyBOK project, not by me, so I left the file as it is. Every chapter the course assigns is now an accessible web page (see CyBOK textbook). The 7 chapters the course does not assign are only in the PDF, which has real, selectable text and a full bookmark outline. The figure descriptions on the web pages are my reading of each figure, and a few small labels in the book's figures are too small to read with certainty.
- **Two labs use visual tools.** Lab 6 has students open a certificate in their own browser, and Lab 8 has students read assembly in Compiler Explorer (godbolt.org). Both are third-party tools I did not build and did not audit. A student who cannot use either one gets an accommodation through the Educational Access Center.

Remaining "needs review" items

axe leaves four kinds of item for a person to judge, and I judged all of them fine:

- **Contrast it cannot compute** (332 groups), where text sits over a layered background like the sidebar or a code block. Since the theme was restyled, axe's AA and AAA contrast rules both flag

the same 166 spots, which is why the count roughly doubled. The by-hand check above covers every one of them at the AAA thresholds.
- **Links with the same name** (10 groups), where a sidebar or outline link and a link in the page share a name and go to the same page.
- **Tables with no data** (4 groups) in Labs 2 and 7. These are blank templates students copy and fill in, so the empty cells are on purpose. Every column has a header.

UDOIT's scan on October 28 found 398 issues, nearly all of them on the new CyBOK pages in Canvas. They are all fixed except the long chapters:

UDOIT issue	Count	Fix
Styles used instead of headings	336	The CyBOK chapters' fourth-level sections are real headings, bold labels run into their sentence, and the meta line at the top of each item and the title of each callout are no longer bold paragraphs
Adjacent links	34	A citation that repeats the one right before it is plain text
Content too long	16	Not fixed, see below
Alt text too long	9	Cut to 125 characters or fewer, with the full description still under "Text description"
Color alone	2, then 55	The "Select Next to continue" bar and the meta line have no fill or text color now, just a blue rule. My first fix set `font-style: normal` on them, and UDOIT's next scan flagged them on 55 items, since it only counts bold or italic as a style
External content	1	The Google Doc on the instructor-only Learning Opportunities page is a link now

Ally's report on October 29 found 20 problems:

Ally issue	Count	Fix
Broken links	2	The “How do I use the Dashboard?” guide moved, so Canvas Resources links to its new home. The instructor-only Learning Opportunities page linked a file in an eCampus course that only its members can open, so it links to the eCampus Faculty Development site instead
Image with no description	1	The home page banner. Ally scores each image file on its own and does not read an SVG’s title, so the description is entered in Ally. A new upload of the banner gets a new file id and needs it again
Image contrast	17	Not fixed, see below

The Dashboard link had moved for a while before Ally caught it. The Instructure Community turns away scripts, so the link check could only list it to open by hand.

The 17 images Ally flags for contrast are CyBOK’s own figures, black text on light gray fills, at about 13:1. I left them as the book publishes them.

The 16 CyBOK chapters run from 7,000 to 37,000 words, and UDOIT suggests splitting any page over 3,000. I kept each chapter whole, because each one is a chapter of the book with its own headings and table of contents, and splitting it would break its footnotes and cross references. The check lists every page over 3,000 words but does not fail on them.